

Joint Detection of AI-Generated Images and Post-Processing Alterations

Deep learning approaches to multi-task classification

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The challenge

AI images detection is often evaluated in **ideal** environments.

Issue: In real life, fake images are often shared multiple times online and thus go through several degradation processes (compression, re-digitization, etc.)

Here we study two **underlying questions**:

- How do different transformations affect Real/fake detection? (Analytical study)
- Can joint detection of the two objectives improve overall performance?

Existing work splits into two camps that rarely meet:

- **Real/fake detection.** The main line of work
 - artifact-based CNNs (Wang et al., 2020)
 - CLIP/foundation-model features (Ojha et al., 2023; Cozzolino et al., 2024)→ Here post-processing is treated as **noise to survive**
- **Manipulation detection.** Models that answer several questions at once
 - ex: detecting and characterizing tampering jointly (Shao et al., 2023).→ This is where our two-head idea comes from, but the task is different (media forgery)

What the field already gives us:

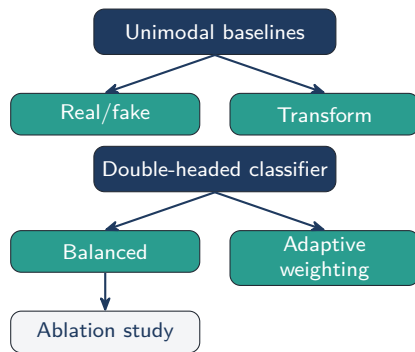
- Multi-task learning framework (shared backbone + multiple heads)
- Robustness to degradation is now a recognized concern (Cozzolino et al., 2024).
- RRDataset formalizes the degradation problem into clean benchmark splits (Li et al., 2025).

Our angle

We flip the transformation from *nuisance* into *target*, and ask the question: do real and fake images react differently to the same post-processing?

The core analysis focuses on three main axis of comparison:

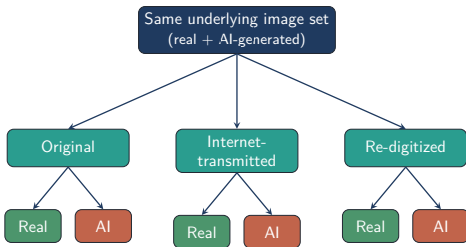
- 1 Comparison **between models** (baseline VS joint detection VS different weighting techniques)
- 2 Comparison in performance between **transform types** within a same model.
- 3 Comparison between image contexts (less important)



Schematic of the architecture

We rely on the **RRDataset** (Li et al., 2025), only the **TEST** set we repurpose.

- Same image set, under three transformation conditions.
- Each condition contains both real and AI-generated images.
- Each image set is divided into different contexts (War, Religion, Politics, etc.)



RRDataset test-set structure

Dataset — examples



Original



Internet-transmitted



Re-digitized

The same image under the three transformation conditions.

Train/test split: We produce a split balanced across **both objectives** + contexts.

	Train	Val.	Test
Unique photos	11 899	2 550	2 550
Individual files	35 697	7 650	7 650
Real	17 847	3 825	3 825
AI-generated	17 850	3 825	3 825
Original	11 899	2 550	2 550
Transfer	11 899	2 550	2 550
Re-digital	11 899	2 550	2 550

Data pre-processing: As we run ResNet as our backbone, we do the following during data loading.

- Standard RGB Normalization
- Apply 244x244 crop on all images: Center crop in val/test, random crop in train
- Apply Random horizontal flip in train

Our experiment is tailored for a **CPU-only** environment.

- 1 ResNet18 as our backbone (lightest "SoA" option)
- 2 we keep the backbone frozen except the last "block" (layer4)
- 3 We replace the original classification head with one or two blank linear layers sharing the same weights.
- 4 in the multitask case, we use a weighted sum of losses:

$$\mathcal{L}_{joint} = \alpha \cdot CE_{realfake} + \beta \cdot CE_{transform}$$

Config & parameters

- batch size: 32
- lr (heads): 1e-3
- lr (layer4): 1e-4
- Adam optimizer
- CE loss

We thus train all models demanded by the guidelines:

- Unimodal baseline on both objectives (10 epochs each)
- Balanced joint detection model ($\alpha = \beta = 1$), also 10 epochs
- Ablation study on weighting: train **3 more epochs** on top of the balanced model with weight ranging from 0.5 to 2. Such implementation was done because of computation time limitations.

We also propose a **small innovation**, introducing a model trained with adaptive weighting. At each epoch, α and β are defined as $1/\text{loss}$ (+ normalized so that $\alpha + \beta = 2$). We train it for 13 epochs.

Validation history confirms all models converge correctly in $\tilde{10}$ epochs (loss < 0.1)

The dataset is perfectly balanced: we choose accuracy as a global, simple metric

Global results:

- A simple finetuned ResNet is already very good on both tasks
- both tasks do not really seem to compete
- Our adaptive model has poor performance (probably too naïve)

Model	R/F	Transf.
Baseline (R/F)	90.68	—
Baseline (transf.)	—	83.10
Joint, balanced	90.61	81.80
Joint, adaptive	90.68	77.56

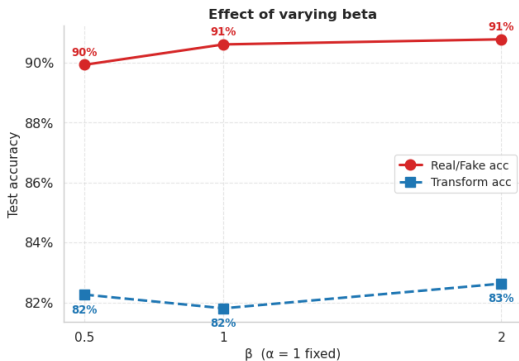
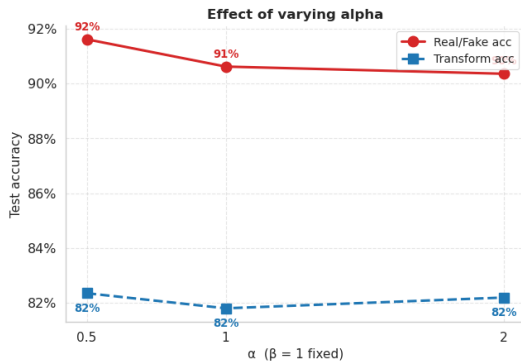
Weighting	Real/Fake	Transform
$\alpha = 0.5, \beta = 1$	91.61%	82.35%
$\alpha = 2, \beta = 1$	90.35%	82.20%
$\alpha = 1, \beta = 0.5$	89.93%	82.26%
$\alpha = 1, \beta = 2$	90.78%	82.63%
$\alpha = 1, \beta = 1$ (balanced)	90.61%	81.80%

Takeaway

Both tasks stay really close (likely due to our training setup). The two tasks again seem to complement: weighting the **weaker** (i.e transform detection) seems to be the way to go. Real/fake seems to hit a cap anyway with aggressive weighting.

Ablation study — visual representation

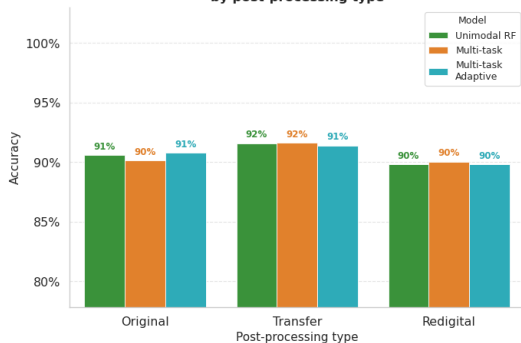
Ablation study: sensitivity to loss weights α and β



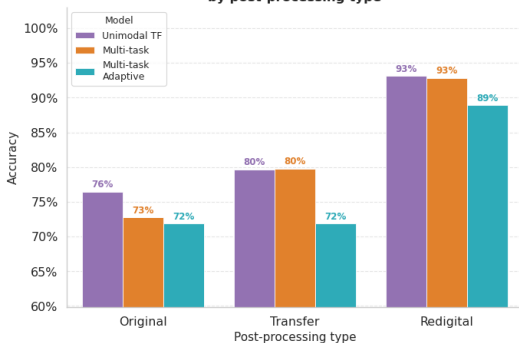
Per-transform analysis

Per-transform performance breakdown

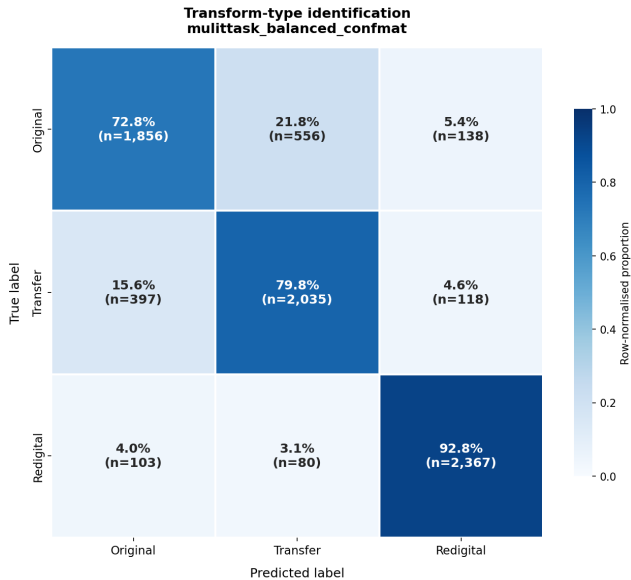
Real/Fake detection accuracy
by post-processing type



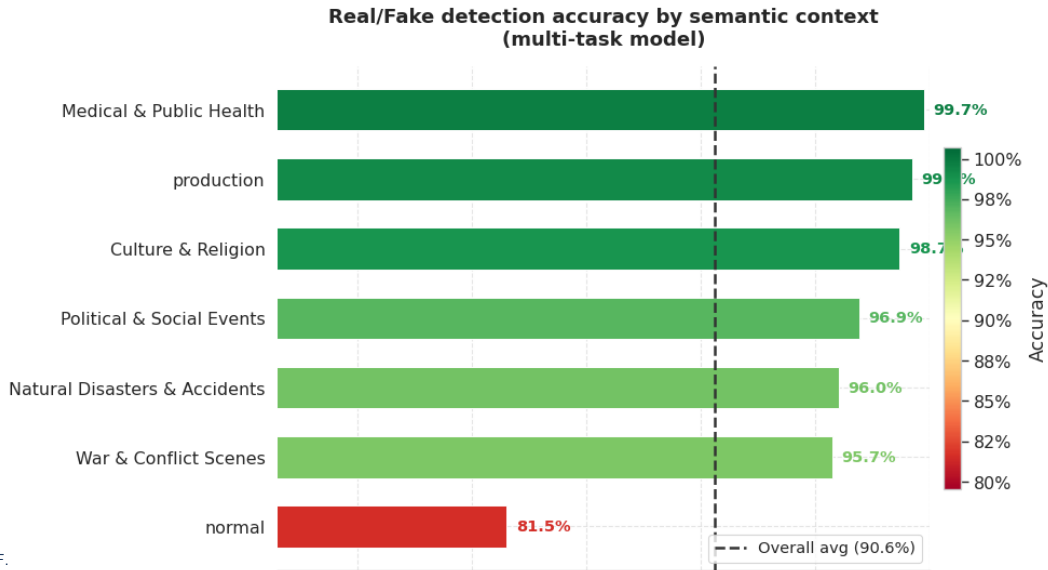
Transform identification accuracy
by post-processing type



Per-transform analysis



Per-context breakdown



Key takeaways

- Joint training nearly matches the unimodal baselines on both tasks => the two tasks **share capacity without competing**.
- Naive adaptive weighting **hurts** transform (77.6%) by under-weighting the harder task.
- Difficulty is **transform-dependent**: re-digitized images are easy to identify but original/transfer are confused.

Future work

- Test more advanced adaptive weighting techniques
- Investigate the weak normal context.
- Conduct a proper ablation study using more computational power.

- C. Li et al. *Bridging the Gap Between Ideal and Real-world Evaluation: Benchmarking AI-Generated Image Detection in Challenging Scenarios*. ICCV, 2025. (RRDataset)
- R. Shao, T. Wu, Z. Liu. *Detecting and Grounding Multi-Modal Media Manipulation*. CVPR, 2023.
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- U. Ojha, Y. Li, Y. J. Lee. *Towards Universal Fake Image Detectors that Generalize Across Generative Models*. CVPR, 2023.
- D. Cozzolino, G. Poggi, R. Corvi, M. Nießner, L. Verdoliva. *Raising the Bar of AI-Generated Image Detection with CLIP*. CVPR Workshops, 2024.
- K. He, X. Zhang, S. Ren, J. Sun. *Deep Residual Learning for Image Recognition*. CVPR, 2016. (ResNet backbone)

Thank you

Questions & discussion